HDOC Day-46

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BATCH-15

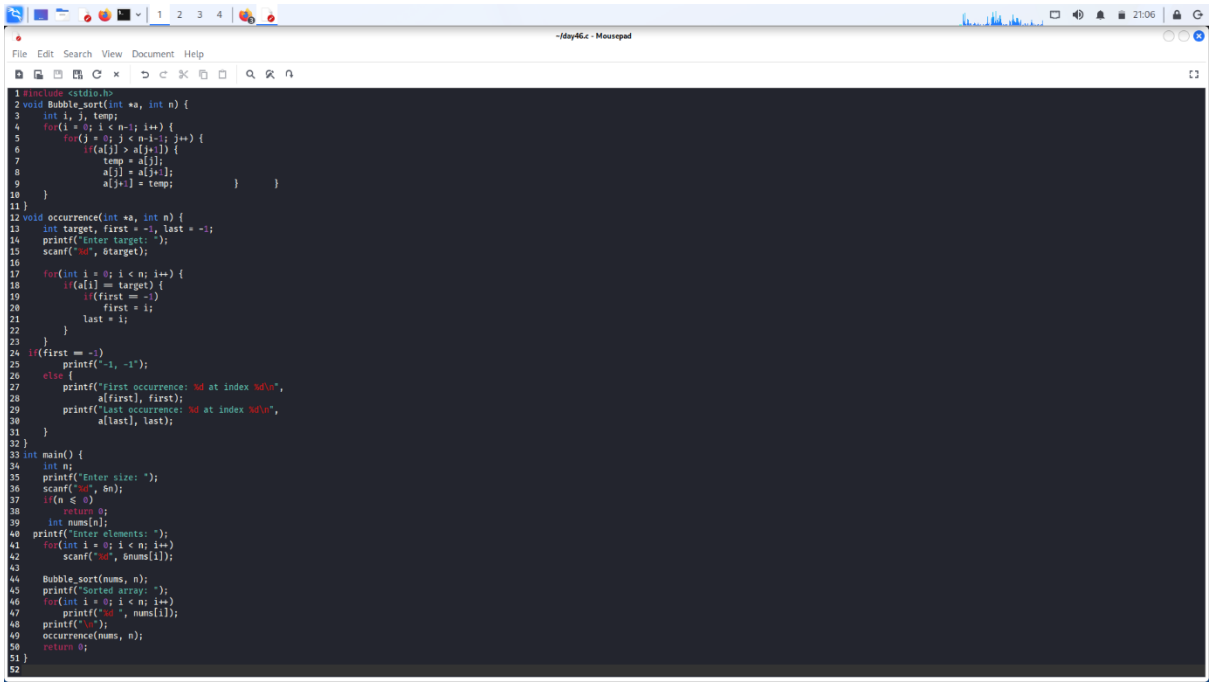
1. Aim

To write a C program to sort an integer array in ascending order using Bubble Sort and find the first and last occurrences of a target element using functions and pointers.

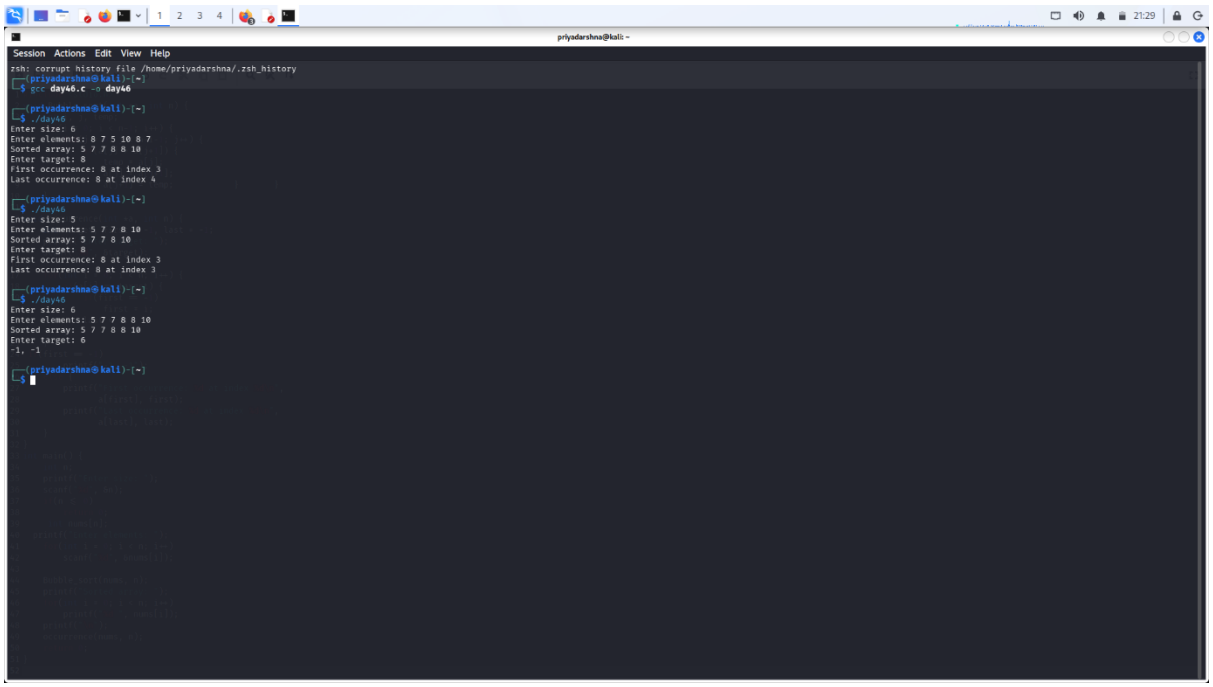
2. Algorithm

1. Start the program.
2. Read the size n of the array.
3. Input n elements.
4. Pass the array to Bubble_sort() using a pointer.
5. Sort the array in ascending order using Bubble Sort.
6. Pass the sorted array to occurrence().
7. Read the target element.
8. Find its first and last indices.
9. Display both occurrences, or -1, -1 if not found.
10. Stop.

>C PROGRAM:


~/day46.c - Mousepad
File Edit Search View Document Help
#include <stdio.h>
void Bubble_sort(int *a, int n) {
int i, j, temp;
for(i = 0; i < n-1; i++) {
for(j = 0; j < n-i-1; j++) {
if(a[j] > a[j+1]) {
temp = a[j];
a[j] = a[j+1];
a[j+1] = temp; } }
}
}
void occurrence(int *a, int n) {
int target, first = -1, last = -1;
printf("Enter target: ");
scanf("%d", &target);
for(int i = 0; i < n; i++) {
if(a[i] == target) {
if(first == -1)
first = i;
last = i;
}
}
if(first == -1)
printf("-1, -1");
else {
printf("First occurrence: %d at index %d\n",
a[first], first);
printf("Last occurrence: %d at index %d\n",
a[last], last);
}
}
int main() {
int n;
printf("Enter size: ");
scanf("%d", &n);
if(n <= 0)
return 0;
int nums[n];
printf("Enter elements: ");
for(int i = 0; i < n; i++)
scanf("%d", &nums[i]);
Bubble_sort(nums, n);
printf("Sorted array: ");
for(int i = 0; i < n; i++)
printf("%d ", nums[i]);
printf("\n");
occurrence(nums, n);
return 0;
}

>OUTPUT


priyadarshna@kali: ~
Session Actions Edit View Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
┌──(priyadarshna㉿kali)-[~]
└─$ gcc day46.c -o day46
┌──(priyadarshna㉿kali)-[~]
└─$ ./day46
Enter size: 6
Enter elements: 8 7 5 10 8 7
Sorted array: 5 7 7 8 8 10
Enter target: 8
First occurrence: 8 at index 3
Last occurrence: 8 at index 4
┌──(priyadarshna㉿kali)-[~]
└─$ ./day46
Enter size: 5
Enter elements: 5 7 7 8 10
Sorted array: 5 7 7 8 10
Enter target: 8
First occurrence: 8 at index 3
Last occurrence: 8 at index 3
┌──(priyadarshna㉿kali)-[~]
└─$ ./day46
Enter size: 6
Enter elements: 5 7 7 8 8 10
Sorted array: 5 7 7 8 8 10
Enter target: 6
-1, -1
┌──(priyadarshna㉿kali)-[~]
└─$